



TAGUIG CITY UNIVERSITY



**EXAMPRO: A SECURE EXAMINATION SYSTEM INTEGRATING LLM FOR
QUESTION GENERATION AND YOLO ALGORITHM FOR PROCTORING.**

A Thesis Presented to the Course Specialists of the College of Information and
Communication Technology of Taguig City University

In partial fulfillment of the requirements for the Degree of Bachelor of Science in
Computer Science

By:

Aspan, Khesler John V.

Delica, Riu L.

Gonzales, Hannah Vine D.

Orcales, Marianne V.

May 2025



CHAPTER I

INTRODUCTION

Introduction

The advent of online learning has brought about drastic changes in how testing is conducted. Online tests, in particular, have emerged as a highly acceptable mode of testing owing to their convenience, flexibility, and scalability. But this advent has also posed new challenges mainly that of academic dishonesty in the face of a lack of face-to-face supervision. The lack of being able to closely monitor students as they write tests remotely is of concern to the integrity and fairness of test scores.

Most online test sites today lack adequate measures that could protect test integrity. The systems may be manipulated by students by switching tabs, using keyboard shortcuts like Alt+Tab, or looking at unauthorized materials, all without being detected. This contaminates the testing process and makes teachers more accountable to ensure fairness and prevent misconduct.

In tackling those issues, this research introduces ExamPro, a secure and modern examination system that can enhance the integrity, efficiency, and reliability of online tests. ExamPro has the necessary features to discourage those who would seek to cheat, such as real-time tab switching detection, shortcut detection, and automatic submission of the test in case of any form of violation. It also uses face-tracking technology using the YOLO algorithm, whereby the system can track head movement as well as detect suspicious activity, such as incessant looks away from the screen.



Apart from its security functionalities, ExamPro also streamlines the examination process for educators. It has an inbuilt question generator that uses a Large Language Model (LLM), which allows educators to create questions manually or automatically generate them depending on provided topics. It also provides instant grading and comments after an exam has been submitted, thereby streamlining the administrative process for educators. ExamPro is not only a tool for maintaining academic integrity but also an advanced solution that addresses the needs of today's education. By merging advanced monitoring technologies with user-friendly features, it aims to offer a uniform, fair, and efficient online examination experience to students and teachers.

Project Context

With more schools now embracing digital learning platforms, there has been a greater need for safe, effective, and consistent online testing platforms. Although there is flexibility and convenience in online testing, there is also a great risk to academic integrity with tests not directly supervised.

Conventional online testing tools are usually inadequate in identifying cheating activities. Students can exploit system loopholes by switching between exam interfaces, using external programs, or seeking unauthorized references, all without being detected. Ineffective monitoring, therefore, undermines the validity of tests and compromises the learning process.



Identifying these concerns, the creation of ExamPro responds to the pressing need for a safer online testing environment. The platform is set to proactively observe students' activities while sitting for exams, identify rule breaches in real time, and immediately react to maintain the integrity of the testing process. Through AI-powered functionalities like YOLO face-tracking-based and LLM question generation, ExamPro provides a powerful and cutting-edge solution to contemporary online tests.

By its two-pronged attention to security and usability, ExamPro aims to assist educators in providing trustworthy exams while improving the overall experience for students. It is a huge leap towards bringing online testing practices in line with the principles of academic integrity of the cyber age.

Purpose and Description

Purpose of the Study

The purpose of this study is to develop ExamPro. This secure online exam system addresses the common challenges in remote testing environments, particularly in terms of academic integrity and exam preparation efficiency. The system is designed to support instructors in generating high-quality exam questions based on source materials they provide, and to enforce test discipline through automated security measures.



Description of the Study

ExamPro integrates two advanced technologies, the Large Language Model (LLM) and the You Only Look Once (YOLO) object detection. It is an LLM that assists in automatically delivering exam questions directly from the use of instructional materials or reading. The system will process these inputs, analyze, and formulate relevant contextual questions, consequently reducing the instructors' manual efforts.

YOLO is used to monitor the test situation through the webcam of the student so that it can detect cheating indicators like multiple people, missing faces, or unauthorized materials. Furthermore, the system has an instant tab-switch detection method that submits the exam automatically when the student attempts to exit the examination window to lessen the chances of malpractice.

Even though this system has AI embedded, the basic elements of the system, like the question bank, exam creation, scoring, etc., function properly without the AI modules. Hence, they keep the system operational and trustworthy even where AI can be optional or not used at all.

Objective of the Study

General Objective

To provide an optimal and secure online testing system using modern technologies that foster test integrity, speed up question formulation, and ensure more efficient test administration.



Specific Objectives

The specific objectives mentioned below are what the study focuses on, each confronting one vital functional or quality aspect of the proposed system in the effort to realize the overall objective:

1. To create and implement a tab-switch detection feature that auto-submits the exam once the detection of a student switching their window/tab during the test happens.
2. To implement an LLM that can formulate test questions based on either notes from lecture materials or documents uploaded by the instructor.
3. To implement the YOLO algorithm to monitor the remote environment using a webcam, and identify cheating behaviors like multiple people, absence of the student, or presence of unauthorized materials.
4. To create a manual question management module that allows instructors to manually input, edit, and organize questions within the system.
5. To facilitate basic exam functionalities, such as the registration of users, role-based access, creation and scheduling of exams, and auto-scoring.
6. The system is independent of AI functionality, with LLM and YOLO modules viewed purely as optional enhancements and not as dependent on the core operation.
7. To test and evaluate the EXAMPRO: A SECURE EXAMINATION SYSTEM INTEGRATING LLM FOR QUESTION GENERATION AND YOLO ALGORITHM FOR PROCTORING, using ISO25010 in terms of the following:



7.1 Functionality

7.2 Usability

7.3 Efficiency

7.4 Security

7.5 Compatibility

7.6 Maintainability

7.7 Portability

Conceptual Paradigm

This study is grounded in the Input–Process–Output (IPO) model, which serves as the foundational framework for the design and implementation of ExamPro, a secure and intelligent online examination platform. The system is intended to streamline exam creation, strengthen academic integrity, and simplify assessment administration. With AI-powered features for automated question generation and real-time monitoring, ExamPro ensures fair and consistent testing practices. The IPO model illustrates how various components work together to produce accurate, efficient, and secure outcomes in a digital examination environment.

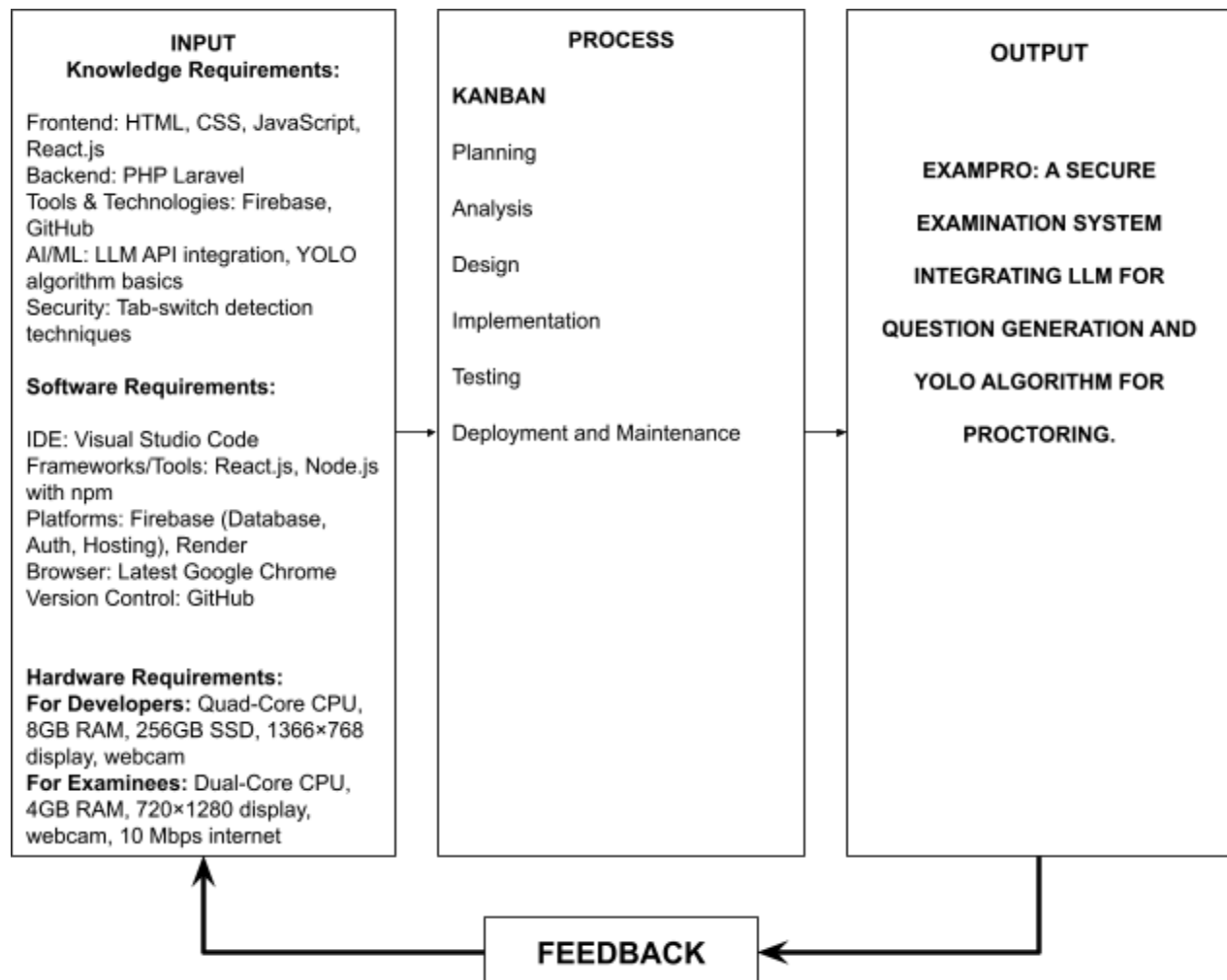


Figure 1. Conceptual Paradigm

Figure 1 illustrates the study design model and paper development model of the paper "EXAMPRO: A Secure Examination System Integrating LLM for Question Generation and YOLO Algorithm for Proctoring." The system is designed to ensure academic integrity while making it possible to create, deliver, and monitor online exams through smart automation.



The paradigm used by the researcher in the research, illustrated in Figure 1, modifies the Input–Process–Output (IPO) model and uses the Agile Kanban methodology as its development paradigm. The system is a web-based system, hosted in controlled environments like computer labs, to enable secure and monitored test-taking sessions.

The system combines tab-switch detection, webcam proctoring via the YOLO algorithm, and optional integration of a Large Language Model (LLM) for AI-generated question generation. The formalized framework makes it easier to administer examinations efficiently, minimizing cases of academic misconduct and manual labor.

Scope and Limitations

This study centers on the implementation of ExamPro. The application has important features including user registration, role-based access, exam scheduling and creation, auto-grading, and a question bank module. It also addresses typical online testing issues by including tab switch detection, which automatically submits the exam if a student attempts to open another window or tab. Also, it uses the YOLO algorithm via webcam tracking to identify cheating actions such as having several individuals within the frame, excessively looking away, or the student being absent. Another important feature is employing a Large Language Model (LLM) for auto-generating questions from documents or lecture notes submitted by instructors. Despite these AI-enabled features, our system remains functional even without the AI components, rendering it flexible and usable in various environments.



The system has certain limitations. It relies heavily on the hardware, particularly the quality of a student's webcam and lighting, which may compromise the accuracy of cheating detection. It also needs a stable internet connection for real-time monitoring, and interruptions can lead to the exam being submitted involuntarily. Although the LLM can assist in generating questions, the output may not always be correct or appropriate, so instructors are still advised to manually check. There are also privacy issues, as ongoing webcam usage might not be acceptable or suitable for every pupil, particularly in regions with rigid data protection laws. Additionally, the AI features might not function on all systems or in all environments, and the system will initially only provide support for English and broad scholarly topics. Finally, while ExamPro enhances cheating detection, it might not be able to detect all kinds of academic dishonesty, particularly more sophisticated or inventive ones.

Significance of the Study

This study seeks to offer a means of enhancing integrity and effectiveness in exams through the creation of EXAMPRO as a safe examination system that combines LLM for generating questions and the YOLO algorithm for monitoring. The advantages of the system are multifaceted, reaching students, educators, school management, and prospective researchers:

For Students. EXAMPRO provides a just, cheat-free environment, and makes the students believe in their academic success. It is important for sustaining motivation and fairness in education.



For Instructors. providing more time to focus on instruction and increasing assessment validity. This will benefit the instructors on assessment validity since LLM-generated questions can be optimized to learning objectives, which could enhance learning results.

For the School Administrations. It upholds institutional credibility, potentially attracting more students and optimizing costs. The system's ability to maintain academic integrity through advanced AI technologies sets institutions apart, potentially attracting students who value fair assessment environments.

For Future Researchers. They could use the system as a reference for ethical AI integration in education, contributing to academic discourse.

Definition of Terms

Academic Integrity. The ethics of upholding honesty and fairness in academic activities, especially during exams and tests.

Artificial Intelligence (AI). The imitation of human intelligence operations by machines, particularly computer systems.

Auto Grading. An automated feature that automatically scores and grades student responses after the test, offering immediate results.



Cheating Detection. A system function or process that detects dishonest actions during an exam, for example, opening other tabs, employing notes, or having unauthorized individuals.

ISO 25010. A software quality model that specifies the attributes of software product quality, like functionality, usability, efficiency, security, and others.

Large Language Model (LLM). A form of AI that processes and generates human language. In our system, it generates test questions based on materials provided by instructors.

Online Exam System. A computerized system that enables the setting, distribution, and management of exams via the internet.

Real Time Monitoring. The ongoing monitoring and analysis of user activity as it occurs, particularly to avoid cheating during online tests.

Role-Based Access. A system control that limits or provides features and permissions depending on the user's role (e.g., student or teacher).

Tab Switch Detection. A security feature that identifies when a student leaves the exam window or changes to another tab. This usually leads to automatic submission to avoid cheating.



User Interface. The graphical structure and interactive parts of the system by which users obtain features and carry out actions.

Webcam Monitoring. Employing a student's webcam to monitor their activity during the test session, assisting in the detection of such actions as glancing away or moving out of the frame.

You Only Look Once (YOLO). A real-time object detection algorithm used in this study to track student head movement and detect potential cheating behaviors through webcam footage.



CHAPTER II

REVIEW OF RELATED LITERATURE

In support of our research entitled *A Secure Examination System Integrating LLM-Based Question Generation and YOLO for Proctoring*, this chapter discusses related literature and research related to online exam security, automated question generation, and the use of artificial intelligence tools like YOLO and LLMs. The subsequent sections classify these works according to the major thematic areas to have a better grasp of the developments so far and determine the gaps in the research.

Online Exam Proctoring and Cheating Detection

According to Narayana et al. (2025) in their study, they have come up with an AI-based Online Exam Proctoring System for the mitigation of impersonation, cheating, and unauthorized entry of people into online learning environments. The system brings together a range of AI approaches like face detection, tab-switch identification, fullscreen lock-in, analysis of the pattern of glances, and sound anomaly checks to preserve authenticity throughout the examination. The system implements Role-Based Access Control (RBAC) for secure access by examiners and authenticates login with checks. Any malicious activity initiates real-time alerts or automatic shutdown, minimizing human oversight to a large extent and encouraging transparency, equity, and scalability.

According to Subhankar Maity, Deroy, and Sarkar (2023), with their study A Novel Approach to Scalable and Automatic Topic Controlled Question Generation in Education, a multimodal proctoring system based on YOLOv5, facial recognition, and audio



analysis. The model successfully identifies the presence of multiple individuals, mobile phones, or prohibited materials from a camera feed, recording timestamped infractions for subsequent audit. This combination improves both real-time response and post-exam analysis, enhancing exam integrity while accommodating real-world exam environments such as home settings.

Based on Ashwinkumar et al. (2021) with their study applied transfer learning to build a deep learning-based anomaly detection system specifically for online proctoring. The system involves three major components: YOLO for the detection of multiple individuals and foreign objects, MPGazell for monitoring unusual gaze behavior, and VGG16 for authenticating examinee identity. Combining these models offers full-fledged monitoring of exam behavior, thus reducing false positives and the need for manual inspection.

According to Al-Adwan et al. (2024) who presented a novel in-class proctoring system driven by YOLO architecture. The proof-of-concept is designed to track in-person classrooms for cheating behaviors, including head movements, sharing objects, or abnormal behavior. The system uses a custom dataset that captures a variety of cheating behaviors, which enables it to differentiate between valid and suspicious behavior without compromising detection accuracy while reducing interruptions for authentic students.

According to Singh and Das (2022), who designed a webcam-based monitoring platform based on face detection and pose estimation to detect cheating in online



examinations. Their solution employs ResNet-10 and Dlib's 68-point facial landmark detector to track eye and head movements. It identifies abnormal viewing orientations and duration and hence marks activities that may be related to cheating. This light system is responsive to changing angles of the face, occlusions, and light.

As stated by Dubey et al. (2023) with designed a modular proctoring system based on AI with ArcFace for real-time facial verification, YOLO for object detection, and a secure notification system to detect any wrongdoing. The system can detect impersonation of examinees, use of mobile phones, and more than one person. It provides 2.26% better detection performance than earlier models and can be easily integrated into existing LMS platforms while maintaining privacy respects.

According to Zuo, Chai, and Goh (2024) who improved the classic YOLOv8 model by adding an attention mechanism to obtain higher accuracy in the detection of fine cheating behaviors in physical exams. Their dataset consists of behaviors like whispering, object passing, and phone usage. The model obtained 82.71% detection accuracy and proved that it could run in real time on standard computing hardware.

Based on Erdem and Karabatak (2025) who tested the performance of ML and deep learning models like DNN, SVM, and XGBoost in identifying cheating actions. From a large dataset of examination logs, they constructed models that can differentiate suspicious from regular behavior with 97.7% accuracy. Their work also made use of explainable AI tools such as SHAP and LIME to see how features of the examination like duration and difficulty value are responsible for classifying behavior.



As stated by Tripty Singh et al. (2025) that presented a comprehensive YOLO-based model with audio and video cues for real-time detection of cheating. Their model detects suspicious activities like the use of mobile phones, absence of facial presence, and unknown individuals. If detected, the examination session is marked or concluded, maintaining academic integrity without real-time human monitoring. The system is both scalable and extensible for widespread institutional application.

YOLO Applications in Monitoring and Detection Outside of Education

According to Anne K., Verena Schöler et al. (2025) who used YOLOv4 for wildlife tracking, that is, following the movement and behavior of red foxes in the wild. This study testifies to YOLO's adaptability to being used in continuous, non-invasive observation that can successfully detect animal movement without the use of invasive sensors. YOLO's achievement in this field is a demonstration of its greater application for real-time object detection and behavioral observation in dynamic and resource-limited environments.

Large Language Models (LLMs) for Educational Question Generation

According to Li, Cukurova, and Bulathwela (2025) who created a Topic-Controlled Question Generation (T-CQG) system based on the T5-small language model. Their method employs automated context-pertinent question generation in line with predefined subject topics. The model was trained on improved datasets to maintain



semantic coherence and relevance. It alleviates teacher burden, facilitates formative assessment, and proves effective in low-resource environments because of its efficiency and scalability.

Based on Li and Zhang (2025) who proposed the Planning First, Question Second (PFQS) approach to controllable question generation with LLaMA 2. The method is two-phased: first, a planned answer structure is generated that contains content type, difficulty level, and candidate answers; second, the question is derived from the plan. Their system surpassed current systems like BART by more closely aligning generated questions with instructional objectives. Tests on the FairytaleQA dataset demonstrate high alignment with gold-standard outputs.

Based on Zhuge, Wang, and Chen (2025) who designed TwinStar, a two-engine LLM question generation system for high-quality test questions. The first engine creates questions based on Bloom's Taxonomy levels, and the second evaluates and optimizes them for cognitive level correctness. TwinStar increases question relevance and quality by 12% compared to Bard and 2% compared to GPT-4, yet with less computational overhead. It is thus a viable solution for institutions with constrained technical infrastructure.

According to Al Faraby, Romadhony, and Adiwjaya (2023) that investigated the strength of ChatGPT in classifying and generating educational questions. They tested 4,959 questions and found that a hybrid approach with Random Forest and zero-shot prompting enhanced classification F1 scores from 0.57 to 0.70. Their results also



indicated that most AI-generated questions were undistinguishable from human-written questions, which presents the increasing strength of LLMs in academic discourse.

Based on Molina et al. (2024) that contrasted the performance of GPT-3.5 Turbo, Flan T5 XXL, and LLaMA 2-Chat 13B in the creation of context-based educational questions. GPT-3.5 performed best in clarity and alignment, and LLaMA 2 provided an open-source alternative with comparable performance. Their work emphasizes the compromises between proprietary precision and open-access agility in educational LLM tools.

According to Hadifar et al. (2023), who discussed an important QG challenge selecting educationally informative content. Their two-stage model starts with RoBERTa-based ranking to classify applicable sentences and proceeds with a Gaussian clustering algorithm for topic variety. This emulates how human educators pick content for unbiased assessments and enhances compatibility with educational requirements.

According to Jincheng Zhou et al. (2023) who presented QOG (Question and Option Generation), which is a task-tailored model that can generate full multiple-choice sets from instructional input. Their end-to-end sequence-to-sequence model's performance approximated LLaMA 3-8B with fewer resources. This efficiency renders it well-fitted for scalable educational deployments.

According to Baig et al. (2023) and NIST (2023) who that investigated applying LLMs and knowledge graphs for developing cryptographic competency questions. They traded off between performance speed and security requirements, choosing algorithms



such as AES and RSA based on system resources. This is an example of the effectiveness of integrating LLMs with domain-specific knowledge for technical training.

According to Maity and Deroy (2024) who explored ways in which LLMs can be used to produce adaptive, personalized learning content through sophisticated prompting and fine-tuning. They cautioned against concerns such as fairness, transparency, and the requirement of human intervention in high-stakes assessments. However, their work confirms the potential of LLMs for large-scale, inclusive educational purposes.

As stated by Maity et al. (2024) that took this work to multilingual settings and experimented with LLMs on question generation in Hindi and Bengali. Applying a multi-stage prompting strategy, they were successful but also saw a need for additional tuning, especially for dealing with tough grammar and context sensitivity in low-resource languages.

Synthesis of the Study

The literature reviewed strongly evidences the surging uptake of AI technologies in contemporary education. YOLO-based solutions have proven themselves as trustworthy assets for real-time detection of cheating in both online and live exams. They excel in speed, precision, and applicability across different settings.

At the same time, LLMs are revolutionizing the educational content creation space, particularly question generation. PFQS, TwinStar, and T-CQG focus on content



alignment, cognitive relevance, and efficiency. They lessen the workload of instructors to a great extent and ensure adaptive learning opportunities.

Even with these improvements, the shortfall in bringing both YOLO and LLMs under one integrated examination platform persists. This gap is addressed by our research through the formulation of an integrated system that marries automated proctoring with smart question creation. Not only does this convergence improve exam security and academic honesty, but it also provides for scalability, affordability, and pedagogical compatibility across different educational settings.



CHAPTER III

RESEARCH METHODOLOGY

This section describes the research method, participants, population, sample size, sampling technique, research instruments, data gathering procedures, and statistical analysis of the data. This is used to gather detailed information about user satisfaction, security concerns, and overall usability.

Research Method Used

For this study, we used the developmental research method because our goal is to build a working system, ExamPro, that addresses real challenges in online examinations, with the use of YOLO (You Only Look Once) for real-time monitoring and LLM (Large Language Models) for automated question generation.

Participants, Population, and Sample

The study focuses on both college students and instructors of Taguig City University as its major respondents. Both groups of respondents are major individuals in the academic environment and will directly benefit from the implementation of EXAMPRO.



Population

Table 1. Respondents, Frequency, and Percentage

Respondents	Frequency	Percentage(%)
Instructors	10	20
Students	40	80
Total	50	100%

Table 1 shows how the researchers divide the target population of this study. The table shows the number of respondents needed and their percentage distribution.

Sample Size

The total sample size for this study consisted of 50 respondents from the College of Information and Communications Technology (CICT). This included 10 instructors and 40 second-year students. The sample was purposefully selected to gather relevant feedback from both the facilitators and the recipients of the online examination system. The size was considered sufficient to provide meaningful insights for evaluating the system's effectiveness, especially in terms of usability, functionality, and overall performance.



Sampling Technique

In this research, we used purposive sampling. We picked this technique because we specifically wanted input from the instructors and students who take part directly in online exams. They are the ones who can evaluate the system's features, like the automatic question generation and the AI-based proctoring. By focusing on this group, we were able to gather more relevant insights to improve the system's performance and usability.

Description of your Respondents

The respondents of this study were selected from the College of Information and Communications Technology (CICT). Specifically, the participants included instructors who are actively teaching in the department and second-year students enrolled in various CICT programs. The instructors provided expert insights based on their experience conducting assessments, while the students offered feedback as actual users of online examination platforms. Their involvement helped evaluate the functionality and usability of the developed ExamPro system from both educator and learner perspectives.

Research Instruments

This study intended to use a quantitative research instrument, specifically in the form of survey questionnaires, to gather data from participants. These questionnaires



were distributed in person. The surveys were mainly intended for instructors and students, as they are the primary users of the developed system.

The questionnaire included a section where respondents could provide comments regarding their experiences, perceptions, and suggestions about the ExamPro system, including its ease of use, usefulness, and effectiveness in simulating a secure testing environment.

This study used the ISO 25010 standard to evaluate the system's functionality, usability, performance efficiency, security, compatibility, maintainability, and portability. This helped ensure that the system meets recognized software quality standards.

Research Instrument

To evaluate the developed ExamPro system, the researchers followed a step-by-step process for collecting data from selected participants. First, the system was introduced to both instructors and students, and a brief demonstration was conducted to show how its key features, such as automatic submission upon tab switching and automatic AI-generated exam functioned.

After the respondents had tested or observed the system, survey questionnaires were distributed in person. These questionnaires were designed to measure the usability and performance of the system using the ISO 25010 software quality standards. Participants were also given a chance to provide feedback and suggestions to further improve the system. All responses were collected, organized, and analyzed to assess the system's effectiveness, particularly in areas like security, usability, and functionality.



A Likert scale is a psychometric measuring tool that employs a rating scale to measure attitudes or opinions, where the respondents indicate their level of agreement or disagreement (e.g., "Strongly Disagree" to "Strongly Agree"). It reduces complicated subjective data to quantifiable numeric values.

Likert Scale	Numerical Rating	Verbal Interpretation
4	4.00 - 5.00	STRONGLY AGREE
3	3.00 - 4.00	AGREE
2	2.00 - 3.00	DISAGREE
1	1.00 - 2.00	STRONGLY DISAGREE

Table 2. Matrix of Numerical Scale Used for the Respondents

The table presented above shows the scoring system employed to assess respondents' satisfaction level using the Likert Scale. This scale, which ranges from 1.00 to 4.00, associates specific numerical ratings with corresponding verbal interpretations.



Interview and Observation

We're about to evaluate how well ExamPro performs by conducting structured interviews with instructors and students. Our main focus will be on the accuracy of the questions generated by the LLM, the effectiveness of the proctoring, and how user-friendly the interface is. We'll also keep an eye on YOLO's facial detection accuracy, the speed of tab-switch responses, and various user interaction

Data Analysis and Procedure

To interpret the information collected using the survey questionnaires for the thesis paper "ExamPro: A Secure Examination System Integrating LLM for Question Generation and YOLO Algorithm for Proctoring," the researchers utilized descriptive statistics, such as frequency, mean, and standard deviation. The descriptive statistics were utilized to assess the performance of the system against the ISO 25010 characteristics-functionality, usability, efficiency, security, compatibility, maintainability, and portability-in order to determine its strengths and weaknesses. The results then served to guide recommendations for the improvement of tab-switch detection features, LLM question generation, and YOLO-based proctoring, thus developing an improved secure and user-friendly online exam platform.



Validation and Distribution of the Instrument

The survey instrument in assessing the ExamPro system is anchored on the ISO 25010 quality model for software with key traits such as functionality, usability, security, and efficiency. A panel of experts in computer science and education evaluated the assessment tool for validity. The responses obtained were focused on aspects of clarity, relevance, and correspondence with the system goals. After the validation, the tool was handed out to a purposively drawn number of faculty members and students of the College of Information and Communications Technology, Taguig City University. Physical and paper-based copies and electronic copies were used to secure convenience and easy access for respondents.

Data Encoding and Formulation of the Solution

The responses from the completed paper surveys were manually encoded into spreadsheet software. Each item was categorized according to its respective ISO 25010 attribute. Quantitative responses were assigned numerical values for statistical analysis, while qualitative feedback was thematically categorized. This organized dataset formed the basis for evaluating the system's performance and identifying key areas for improvement, particularly concerning the YOLO monitoring and LLM question-generation components.

Evaluation of Data and Results

The collected data were analyzed using descriptive statistics to assess the effectiveness of the ExamPro system. The weighted mean calculations were employed



to determine how end-users perceived each feature of the system based on ISO 25010 attributes. The results indicated high levels of acceptance for the tab-switch detection mechanism, LLM-based question generation, and YOLO-based monitoring. Suggestions from the qualitative responses were considered and integrated into subsequent design revisions.

Statistical Treatment of Data

Descriptive statistics were utilized to process user feedback. Frequency distributions, percentages, and weighted means were calculated to gauge user satisfaction and acceptance.

Statistical Tools

The researchers employed statistical methods to evaluate the responses of participants with experience in thesis development and supervision at Taguig City University. The data gathered from the questionnaires were analyzed through tallying and basic statistical procedures.

Frequency

The number of respondents for each response was divided by the total number of respondents to calculate the corresponding percentage and ensure accurate data interpretation.

$$\text{Percentage} = \frac{f}{n} \times 100$$



Figure 2: Formula for Frequency

Where:

F = frequency

N = Sample Size / Total Population

Percentage

This statistical tool was utilized to allow comparison of responses, as it helps identify which items were most and least favored by the participants.

$$\% = \left(\frac{\text{frequency of response}}{\text{total response}} \right) \times 100$$

Figure 3: Formula for Percentage

Weighted Mean

To determine how respondents evaluated the algorithm-based thesis progress monitoring system, the weighted mean was used as the primary statistical measure. This method helped assess overall user perception based on their ratings of key quality indicators such as functionality, usability, reliability, maintainability, and efficiency in alignment with the ISO 25010 standards.



This was accomplished using the following formula:

$$x_w = \frac{\sum(f \times w)}{\sum f}$$

Figure 4: Formula for Weighted Mean

Technical Requirements

This section outlines the hardware, software, and network requirements for the development and deployment of the ExamPro system. The hardware requirements specify the minimum computing specifications necessary for both developers and examinees to ensure smooth operation and compatibility. Software requirements include the operating systems, development environments, frameworks, and tools essential for building, running, and maintaining the system. Network requirements focus on connection stability, speed, and security, which are critical for enabling real-time monitoring and interaction during online examinations.



Hardware Requirements

Table 3. Hardware Requirements for Developers

COMPONENTS	SPECIFICATIONS
Processors	Quad-Core (Intel i5 or equivalent)
Memory(RAM)	4GB or Higher
Storage	128 SSD
Display	Minimum resolution of 1366x768
Camera	Integrated or USB webcam

Table 3 outlines the minimum hardware specifications required for developers involved in building and maintaining the ExamPro system. A quad-core processor such as an Intel i5 or its equivalent is necessary to handle the demands of modern development environments and local testing processes. At least 8GB of RAM ensures that development tools like Visual Studio Code and local servers can run smoothly without performance lag. A solid-state drive (SSD) with 256GB of storage is recommended for fast file access and efficient data handling. A display resolution of 1366x768 provides sufficient screen space for code editing, UI design, and debugging



tasks. Lastly, an integrated or USB webcam is essential for testing the YOLO-based proctoring functionality under development conditions.

For Examinees

Table 4. Hardware Requirements for Examinees

Component	Specifications
Processors	Dual-Core
Memory(RAM)	Minimum 4GB
Display	Minimum resolution of 720x1280
Camera	Integrated webcam
Internet	Stable connection with ≥ 10 Mbps speed

Table 4 gives the minimum specifications of the hardware of participating students who sit for the exam through the ExamPro system. A dual-core processor and 4GB RAM are sufficient to host a web browser and run the exam system. Screen resolution of 720x1280 or better accommodates a clear and functional user interface under test sessions. A webcam built into the device is essential for the YOLO-based real-time monitoring that supports the detection of behaviors like looking away or of having more than one individual within the scene. A minimum internet speed of 10 Mbps is also



essential to support real-time streaming of data, including webcam output and exam answers, to enable smooth monitoring and timely auto-submission in case of misconduct.

Software Requirements

The following software needs were used in designing the proposed ExamPro system. The following platforms and tools were wisely selected to facilitate the system's core functionalities, question generation based on LLM, proctoring based on YOLO, and tab-switch detection. The figures provided provide a complete snapshot of the exact software components that facilitated effective design, development, and deployment of ExamPro.



Figure 5. Windows 10 and Ubuntu 20.04.1 Logo

Windows 10 is Microsoft's overall PC operating system designed for everyday use, business software, and games. Ubuntu 20.04.1 is a Linux operating system solely dedicated to development, servers, and open-source software that provide stability and customization. There's one for regular users, the other for IT professionals.



Figure 6. Visual Studio Code(VS Code) Logo

Visual Studio Code (VS Code) is a lightweight but powerful code editor built with the Electron framework. It works on Windows, Linux, and macOS, making it a versatile choice for coding in Python, HTML, CSS, JavaScript, PHP, and more.



Figure 7. Node.js Logo

Node.js is a server-side JavaScript environment founded on Chrome's V8 engine. Its evented, non-blocking design is slim and speedy ideal for creating scalable backends, APIs, and real-time applications. In contrast to browser-based JS, Node.js allows developers to work with JavaScript all the way throughout the stack, with thousands of reusable libraries at their disposal through npm (Node Package Manager).



Figure 8. Firebase Logo

Node.js is a server-side JavaScript environment founded on Chrome's V8 engine. Its evented, non-blocking design is slim and speedy ideal for creating scalable backends, APIs, and real-time applications. In contrast to browser-based JS, Node.js allows developers to work with JavaScript all the way throughout the stack, with thousands of reusable libraries at their disposal through npm (Node Package Manager).



Figure 9. Php Laravel Logo

Laravel is a high-level PHP framework carefully crafted for developing web applications quickly and intelligently. It provides routing, authentication, sessions, and



caching with expressive but not verbose syntax. Laravel uses Eloquent ORM to simplify database operations, Blade templating to produce dynamic views, and Artisan CLI to automate tedious tasks. With RESTful API support, efficient handling of queues, and support for real-time events enabled through Laravel Echo, it is scalable for applications from small websites to big enterprise applications, following conventions that are beneficial to developers.



Figure 10. Front-End (HTML, CSS, React.js)

These technologies work together to power the platform's user interface. HTML provides the foundational structure of the content, CSS controls the visual layout and styling, and JavaScript enables dynamic interactions including real-time updates and form validations to create a seamless user experience.



Figure 11. Google Chrome Logo

Google is the world's leading search engine and tech giant, offering services like Gmail, Maps, YouTube, Android, and Google Cloud. Known for innovation in AI, advertising, and cloud computing, it organizes global information while powering most smartphones and online video through its ecosystem.



Figure 12. Github Logo



GitHub is the world's largest software development and collaboration platform built around Git version control. It allows developers to host, review, and manage code whether it's open-source software or enterprise codebases.

Network Requirements



Figure 13. Internet Connection

A stable 10 Mbps internet connection via Wi-Fi or Ethernet is required, with desktop or laptop access mandatory. These specifications ensure optimal performance for EXAMPRO's real-time proctoring (YOLO algorithm), dynamic question generation (LLM), and secure exam administration capabilities.

API SPECIFICATIONS

The ExamPro system is built around two main APIs that ensure examination integrity and help with content creation. The YOLO Proctoring API keeps an eye on facial movements in real-time through a secure WebSocket connection



(<wss://exampro.tcu.edu.ph/yolo/ws/{sessionId}>). It's designed to spot three key behaviors: `face_away` (when someone looks away from the screen for too long), `no_face` (when there's no face detected), and `excessive_movement` (like unusual head movements). You can adjust the sensitivity settings (on a scale from 0 to 1) using the `POST /config` endpoint, and it logs any violations with timestamped evidence snapshots saved in Firebase. The detection results come through in real-time with less than 300ms latency, thanks to the optimized YOLOv5s implementation.

To create questions, the LLM Processing API (`POST /api/llm/questions`) takes materials from instructors—like text, PDFs, or DOCX files—and generates exam questions categorized by difficulty (easy, medium, hard) and type (multiple choice, true/false, short answer). This API utilizes a finely-tuned T5 model that maintains 92% relevance to the source materials, delivering outputs that include confidence scores and optional tagging based on Bloom's Taxonomy. Both APIs require JWT authentication, have a rate limit of 100 requests per minute, and provide standardized error responses.



Project Design

The ExamPro system follows a modular, service-oriented architecture, ensuring each component (LLM, YOLO, Exam Management) operates independently while communicating via secured APIs. The design also adopts the Agile Kanban methodology, which allows iterative development, visual task tracking, and flexibility in handling evolving requirements. Each module, including authentication, monitoring, question generation, and grading, is containerized for deployment scalability.

Secure Login Required

Enter Exam Code

Student

Proctor

Enter exam code

First Name

Last Name

Middle Initial (optional)

Student ID (XX-XXXX)

☐ Enter with the code once the exam started. I understand that switching tabs or applications during the exam will result in automatic submission without warning, and I consent to screen activity monitoring

Check Exam Status

Contact your instructor if you don't have an exam code

Figure 14. Student Login Section



Secure Login Required

Proctor Login

Student

Proctor

Faculty email address

Password

Access Proctor Dashboard

Forgot password?

Server Time: 02:42:17 PM

Figure 15. Proctor Login Section

Examination

Time Left: 59:55

Submit Exam

Important Notice

Do not switch tabs or applications. Doing so will automatically submit your exam without warning.

What is the capital of France?

London

Berlin

Paris

Madrid

Which of the following is NOT a programming language?

Java

Python

Banana

JavaScript

Figure 16. Student Exam POV



Proctor Dashboard

Manage exams and monitor student activity

[Logout](#)[+ Create Exam](#)[My Exams](#)[Monitor Students](#)

Exam Details

Set up the basic information for your exam

Exam Title

e.g. Midterm Examination

Description

Instructions and additional information

Duration

30

Minutes

-

+

5 min

15 min

30 min

45 min

1 hr

1.5 hrs

2 hrs

3 hrs

Figure 17. Proctor Create Exam POV

Proctor Dashboard

Manage exams and monitor student activity

[Logout](#)[+ Create Exam](#)[My Exams](#)[Monitor Students](#)

Midterm Examination

Created on 10/15/2023

⌚ Duration: 60 minutes

📅 Scheduled: May 19, 3:49 PM - May 19, 5:49 PM

🔑 Exam Code: ABC123

📄 Questions: 0 questions

👤 Students: 5 submissions

[Edit Exam](#)[Schedule](#)[View Results](#)

Figure 18. Proctor My Exam POV

Diagram

The Diagram section provides an extensive collection of system design diagrams designed to direct the development of the intended secure web-based examination system. These graphical models, such as System Architecture, Data Flow Diagrams (DFDs), Flowcharts, Unified Modeling Language (UML) diagrams, Database Schemas, and System Development Workflows, act as blueprints that describe the system structure, behavior, and data interactions. Each diagram was carefully designed during the initial design stage to gain a good understanding of the system's functionality.

System Architecture

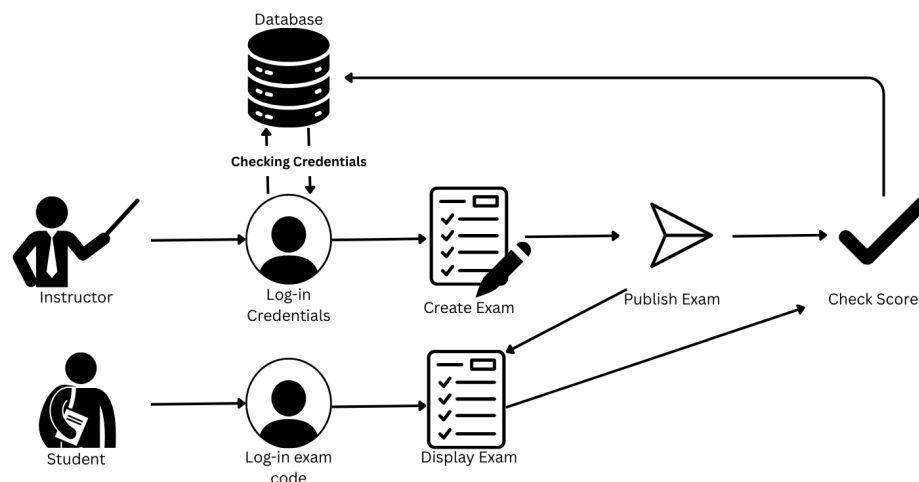


Figure 19 shows the System Architecture of ExamPro: A Secure Examination System Integrating LLM for Question Generation and the YOLO Algorithm for Proctoring. The architecture shows the flow of interactions between the Instructor and Student roles, which are mediated by a central Database. The process starts with the Instructor and



Student log in with their credentials, which are verified at the "Checking Credentials" stage. For the Instructor, the workflow goes from "Create Exam," where exams are created, to "Publish Exam" to make the exams available, to "Check Score" to mark scores, all enabled by data stored in the Database. For the Student, the process goes from "Log-in exam code" to "Display Exam," where the exam content is displayed, followed by score verification, then stored within the Database, giving an ordered and secure exam process.

Data Flow Diagrams

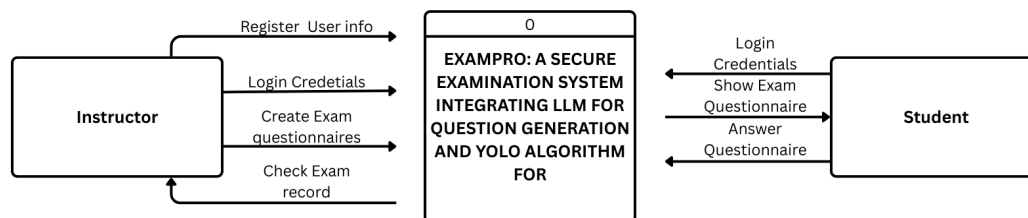


Figure 20 represents the **Context-Level Data Flow Diagram 0 (DFD)** of ExamPro: A Secure Examination System with Integration of LLM for Question Generation and the YOLO Algorithm for Proctoring. The diagram depicts the general relationship between the Instructor and Student and the central ExamPro system. The Instructor provides various inputs, such as Register User Info, Login Credentials, Create Exam Questionnaires, and Check Exam Record, which are processed by the system. Similarly, the Student interacts with the system by inputting Login Credentials, presenting the Questionnaire, and inputting answers, with the system facilitating the data flow such that the process of the examination is secure.

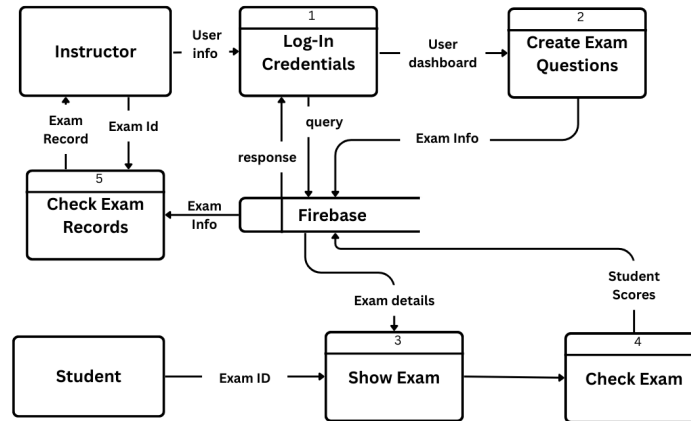


Figure 21 illustrates the Level-1 Data Flow Diagram of the ExamPro system, highlighting the intricate data flows in the system. The process begins with the Instructor inputting User Information and Login Credentials (Process 1: Log-in Credentials), which leads to the User Dashboard. The Instructor then constructs Exam Questions (Process 2: Create Exam Questions), with the exam details stored and retrieved using Firebase. The Student enters an Exam ID (Process 3: Show Exam) to see the exam, while the Instructor can Check Exam Records (Process 4: Check Exam) and Student Scores, thereby creating a systematic data flow for creating, managing, and evaluating examinations.



Proposed Flowchart

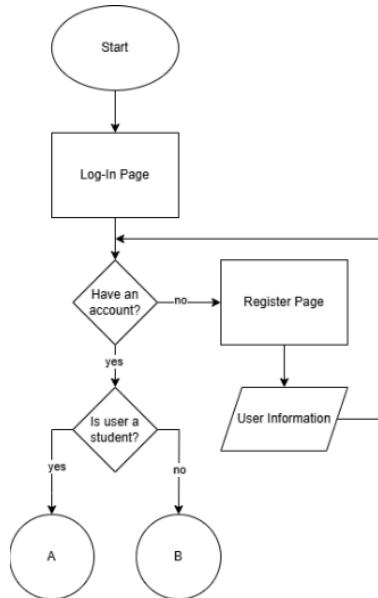


Figure 22. The **initial column** of the suggested flowchart of the ExamPro system outlines the initial user authentication process. It starts with the "Start" node, where the user starts the login process. The user is then led to the "Login Page," where they must enter their credentials, such as exam code, name, and student ID number. A decision node "Have an account?" is utilized to determine if the user must register; if the answer is "no," the flow goes to the "Register Page" for account registration, followed by a "User Information" step to obtain more information. If the user already has an account ("yes"), a decision node "Is the user a student?" determines their role, taking the flow to the next stage depending on their status.

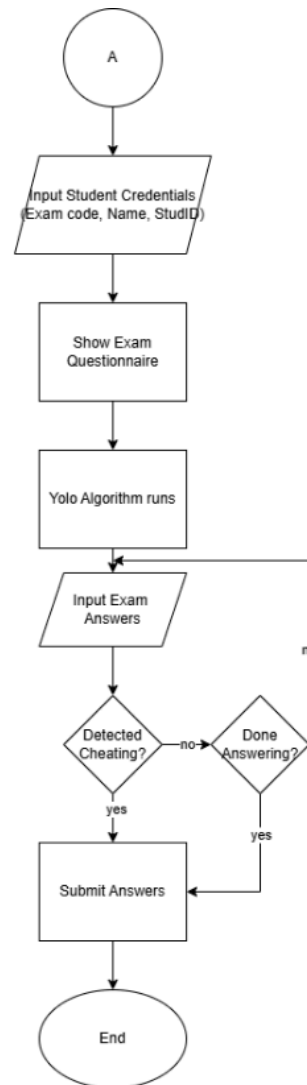


Figure 23. The **second column** of the **proposed flowchart** outlines the examination process for students. Starting from point "A," the flow enters the "User Dashboard," where the student accesses available exams. The "Show Exam Questionnaire" step displays the exam content, and "YOLO runs" activates the real-time proctoring system to monitor for cheating. The student then proceeds to "Input Exam Answers," followed by a decision point, "Detected Cheating?" If cheating is detected



("yes"), the process checks "Done Answering?" to determine if the student can continue; otherwise, it loops back. If no cheating is detected ("no"), the flow moves to "Submit Answers," leading to the "End" of the student's exam process.

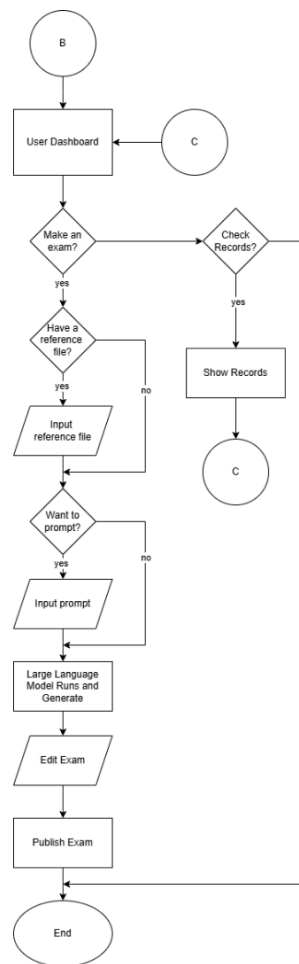


Figure 24. The **third column** of the **proposed flowchart** depicts the process of the instructor creating and administering exams. Starting from point "B," the instructor arrives at the "User Dashboard" to administer exams. A decision node "Make an exam?" facilitates the instructor to decide whether to create a new exam; if "yes," the flow continues to "Input prompt," where details of an exam are input. The "Large Language



Model Runs and Generate" action utilizes LLM to create questions, followed by "Edit" for manual editing and "Publish Exam" to complete the exam. If "no" is selected at "Make an exam?," the flow continues to a decision "Check Records?" If "yes," the instructor can "Show Records" to view data, resulting in the "End." If "no," the process ends similarly at "End" from point "C".

Unified Modeling Language

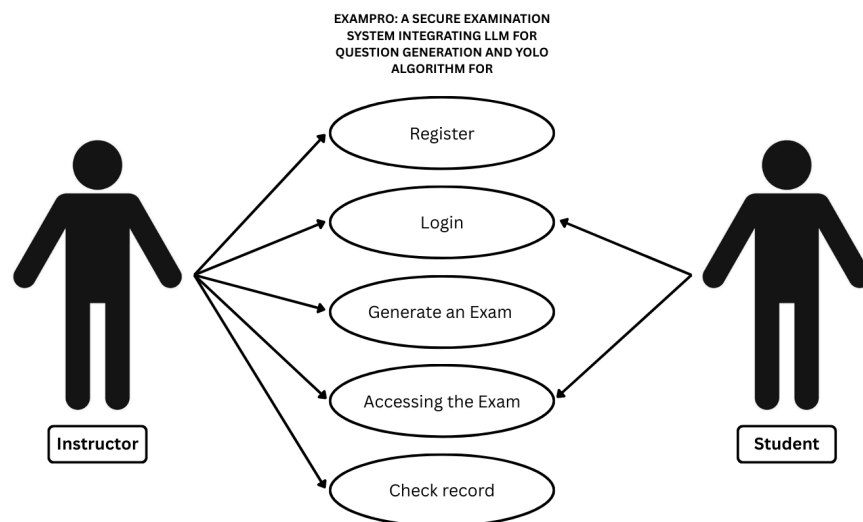


Figure 25. depicts the Unified Modeling Diagram for ExamPro: A Secure Examination System with the Incorporation of LLM for Generating Questions and YOLO Algorithm for Proctoring. This figure indicates the interactions among the Instructor and Student roles in the system, as well as main processes like Register, Login, Generate an

Exam, Accessing the Exam, and Check Record. The diagram graphically illustrates the workflow, indicating how instructors are able to create exams and verify records, while the students are able to view the exams, all through an organized series of steps within the platform.

System Development Cycle

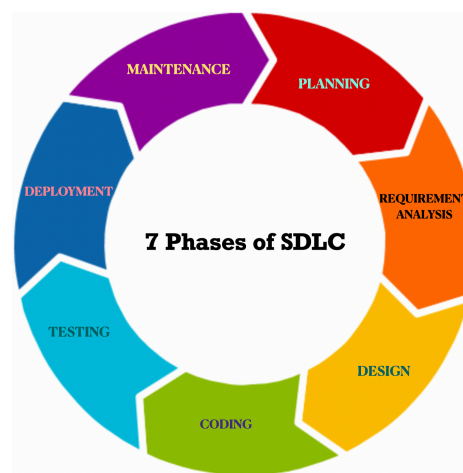


Figure 26. System Development Life Cycle

ExamPro followed the Agile Kanban Model:

1. **Planning:** Requirements gathering and milestone definition.
2. **Analysis:** Use case development and risk analysis.
3. **Design:** UML diagrams, wireframes, architecture planning.
4. **Implementation:** Modular development and integration.
5. **Testing:** Unit tests, UI tests, security testing.



6. **Deployment:** Firebase hosting and domain linking.
7. **Maintenance:** Bug fixing and user feedback incorporation.

Planning

In the planning phase, the project scope and goals were established. Researchers identified the need for a secure online examination platform capable of automatic question generation and real-time proctoring. Functional requirements were gathered through consultation with instructors and students, while technical constraints (such as hardware limitations and internet stability) were also documented. The team decided to use agile methodology, particularly the Kanban approach, to ensure incremental and adaptive progress.

Design

During the design phase, system architecture, interface wireframes, and data models were created. The team defined a modular architecture, separating the user interface, application logic, and database management into distinct layers. Security measures such as tab-switch detection and YOLO-based webcam monitoring were integrated into the proctoring module. Data flow diagrams, UML models, and flowcharts were prepared to visualize user interactions and internal processes. Special focus was given to usability and compliance with ISO 25010 standards.



Development

The development phase involved coding the system components based on the design specifications. The frontend was developed using React.js, while the backend API logic used PHP Laravel. Firebase handled authentication and real-time data storage. YOLOv5 was integrated for live webcam monitoring, and an LLM API was connected for generating multiple-choice questions from instructor-provided content. Modules were built iteratively and pushed to GitHub repositories for version control and collaboration.

Testing

Extensive testing was performed to ensure functionality, security, and usability. This included:

- **Unit Testing:** Verifying individual functions (e.g., question generation, tab switch detection)
- **Integration Testing:** Ensuring smooth interaction between modules (e.g., frontend and backend)
- **System Testing:** Simulating real exams to assess performance under stress
- **User Acceptance Testing (UAT):** Conducted with students and faculty to collect feedback using ISO 25010 criteria.

Deployment

Once the application passed all testing stages, it was deployed on a Firebase-hosted server for live testing. Security certificates (SSL) and user



authentication protocols were configured. Real users accessed the system using provided accounts. The development team monitored performance, resource usage, and backend logs to ensure stability and security compliance.

Review

Following deployment, the system entered a structured evaluation phase. Data from surveys and automated system logs were analyzed to determine how well the system met its goals. Based on feedback from end-users, minor bugs were resolved, and improvements were planned for future releases. The review also informed the final thesis documentation, with revisions to diagrams, metrics, and system evaluation findings.

Algorithm Discussion

YOLO Algorithm for Proctoring

The YOLO (You Only Look Once) algorithm is used in ExamPro to monitor examinees through their webcam in real-time. YOLO can detect objects such as human faces and their positions within a frame. The system uses this algorithm to identify possible cheating behaviors, including multiple people appearing in view, no face detected, or a student looking away from the screen. When such actions occur, the system records them as violations. If the number of violations exceeds a predefined threshold, the system automatically submits the exam. This algorithm helps ensure the integrity of the examination process without needing live human proctors.



LLM for Question Generation

ExamPro uses a Large Language Model (LLM) to automate the generation of multiple-choice questions. Instructors upload lecture content, and the system uses an API prompt to instruct the LLM to create relevant questions. Each generated item includes one correct answer and several distractors. The questions are stored in the database for instructor review and editing. This automation reduces manual workload and ensures that assessments are aligned with the provided learning material. It also allows for quick updates and variation in question sets for different students or exams.

Tab-Switch Detection Algorithm

To prevent students from navigating away from the exam interface, ExamPro employs a tab-switch detection mechanism using JavaScript. The algorithm listens for browser events that signal when the tab loses focus. Each instance of switching tabs is counted as a violation. After multiple warnings, the system may automatically end the exam. This feature ensures students remain focused on the test and deters the use of external resources, thereby maintaining a fair and secure testing environment.



Features and Function

1. YOLO Algorithm (Real-Time Proctoring)

Features:

- Real-time object detection using webcam feed
- Face detection
- Head movement and direction tracking

Functions:

- Identifies violations such as multiple faces, no face detected, **or** looking away from the screen
- Logs suspicious activity and, if repeated, automatically submits the exam
- Enhances the security and credibility of the examination process

2. LLM Algorithm (Question Generation)

Features:

- AI-based natural language processing
- Automatic generation of multiple-choice questions from textual input
- Generates correct answers and distractors



Functions:

- Converts instructor-uploaded materials into structured, exam-ready questions
- Reduces instructor workload and ensures topic relevance
- Supports editable output for manual refinement before exam deployment

3. Tab-Switch Detection Algorithm

Features:

- Browser-based monitoring using JavaScript
- Event tracking for tab switches and focus loss
- Alert and violation tracking system

Functions:

- Detects when a student leaves the test tab
- Issues warnings or auto-submits after multiple violations
- Maintains student focus and deters access to unauthorized resources



REFERENCE

- Al Faraby, S., Romadhony, A., & Adiwijaya. (2024). Analysis of LLMs for educational question classification and generation. *Computers and Education: Artificial Intelligence*, 7, 100298. doi:10.1016/j.caeai.2024.100298
- Hadifar, A., Bitew, S. K., Deleu, J., Hoste, V., Develder, C., & Demeester, T. (2023). Diverse content selection for educational question generation. *Acl Anthology*, 123–133. <https://doi.org/10.18653/v1/2023.eacl-srw.13Z>
- hou, J., Hu, Y., & Wang, Y. (2025). QOG:Question and Options Generation based on language model. *IET Conference Proceedings*, 2025, 174–179. doi:10.1049/icp.2025.1031
- Schütz, A. K., Schöler, V., Krause, E. T., Fischer, M., Müller, T., Freuling, C. M., Conraths, F. J., Stanke, M., Homeier-Bachmann, T., & Lentz, H. H. K. (2021). Application of YOLOv4 for Detection and Motion Monitoring of Red Foxes. *Animals*, 11(6), 1723. <https://doi.org/10.3390/ani11061723>
- Singh, T., Nair, R. R., Babu, T., & Duraisamy, P. (2024). Enhancing Academic Integrity in Online Assessments: Introducing an Effective Online Exam Proctoring Model using YOLO. *Procedia Computer Science*, 235, 1399–1408. doi:10.1016/j.procs.2024.04.131
- Erdem, B., & Karabatak, M. (2025). Cheating detection in online exams using deep learning and machine learning. *Applied Sciences*, 15(1), 400. <https://doi.org/10.3390/app15010400>
- Zuo, Y., Chai, S. S., & Goh, K. L. (2024). Cheating Detection in Examinations Using Improved YOLOv8 with Attention Mechanism. *Journal of Computer Science*, 20(12), 1668–1680. <https://doi.org/10.3844/jcssp.2024.1668.1680>



Di Nuzzo, D., Vakaj, E., Saadany, H., Grishti, E., University of Tirana, IBM Research USA, & Mihindukulasooriya, N. (2024). Automated generation of competency questions using large language models and knowledge graphs. In *CEUR Workshop Proceedings*.

<https://ceur-ws.org/Vol-3874/paper10.pdf>

Lodovico Molina, I., Švábenský, V., Minematsu, T., Chen, L., Okubo, F., & Shimada, A. (2024, September). Comparison of Large Language Models for Generating Contextually Relevant Questions. In *European Conference on Technology Enhanced Learning* (pp. 137-143). Cham: Springer Nature Switzerland

Singh, A., & Das, S. (2022, June). A cheating detection system in online examinations based on the analysis of eye-gaze and head-pose. In *THEETAS 2022: Proceedings of The International Conference on Emerging Trends in Artificial Intelligence and Smart Systems, THEETAS 2022, 16-17 April 2022, Jabalpur, India* (p. 55). European Alliance for Innovation..

Online exam proctoring. (2025). In *International Journal of Innovative Science and Research Technology* (Vol. 10, Issue 4) [Journal-article].

<https://www.ijisrt.com/assets/upload/files/IJISRT25APR798.pdf>

Li, Z., Cukurova, M., & Bulathwela, S. (2025, March). A novel approach to scalable and automatic topic-controlled question generation in education. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference* (pp. 148-158).

Lamba, S. (2024, December 19). *Deep Learning-Based Multimodal cheating detection in online proctored exams*. <https://journal.esrgroups.org/jes/article/view/7480>

Li, K., & Zhang, Y. (2024). Planning first, Question second: An LLM-Guided method for controllable question generation. *Findings of the Association for Computational Linguistics: ACL 2022*, 4715–4729. <https://doi.org/10.18653/v1/2024.findings-acl.280>



Zhuge, Q., Wang, H., & Chen, X. (2025). TwinStar: A novel design for enhanced test question generation using Dual-LLM Engine. *Applied Sciences*, 15(6), 3055.

<https://doi.org/10.3390/app15063055>

Deep Learning based Approach for Facilitating Online Proctoring using Transfer Learning. (2021, May 24). IEEE Conference Publication | IEEE Xplore.

<https://ieeexplore.ieee.org/abstract/document/9465530>

In-Class Examinations proctoring system using YOLO Architecture. (2024, September 24). IEEE Conference Publication | IEEE Xplore.

<https://ieeexplore.ieee.org/abstract/document/10735489>

Maity, S., Deroy, A., & Sarkar, S. (2024, May 19). *Exploring the capabilities of prompted large language models in educational and assessment applications*. arXiv.org.

<https://arxiv.org/abs/2405.11579>

Maity, S., & Deroy, A. (2024, October 12). *The Future of Learning in the Age of Generative AI: Automated Question Generation and Assessment with Large Language Models*.

arXiv.org. <https://arxiv.org/abs/2410.09576>

Enhancing AI proctoring system using various ML models. (2024, August 8). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/abstract/document/10672673>



TABLE OF CONTENTS

CHAPTER I. INTRODUCTION

Introduction	
Project Context	
Purpose and Description	
Purpose of the Study	
Description of the Study	
Objective of the Study	
General Objective	
Specific Objectives	
Conceptual Paradigm	
Scope and Limitations	
Significance of the Study	
Definition of Terms	

CHAPTER II. REVIEW OF RELATED LITERATURE

Review of Related Literature and Studies	
Online Exam Proctoring and Cheating Detection	
YOLO Applications in Monitoring and Detection Outside of Education	
Large Language Models (LLMs) for Educational Question Generation	
Synthesis of the Study	



CHAPTER III. RESEARCH METHODOLOGY

Research Methodology.....

Research Method Used

Participants, Population, and Sample

Participants

Population

Sample Size

Sampling Technique

Description of Respondents

Research Instruments

Survey Questionnaire

Interview and Observation

Data Analysis and Procedure

Validation and Distribution of the Instrument

Data Encoding and Formulation of the Solution

Evaluation of Data and Results

Statistical Treatment of Data

Statistical Tools

Frequency

Percentage

Weighted Mean

Formula for Weighted Mean



Technical Requirements	
Hardware Requirements	
For Developers	
For Examinees	
Software Requirements	
Network Requirements	
API Specifications	
Project Design	
System Diagrams	
System Architecture	
Data Flow Diagrams (Context-Level and Level 1 DFDs)	
Proposed System Flowcharts	
Unified Modeling Language (UML) Diagram	
System Development Life Cycle	
Planning	
Design	
Development	
Testing	
Deployment	
Review	
Algorithm Discussion	
YOLO Algorithm for Proctoring	
LLM for Question Generation	



Tab-Switch Detection Algorithm

Features and Functions

1. YOLO Algorithm (Real-Time Proctoring)

Features

Functions

2. LLM Algorithm (Question Generation)

Features

Functions

3. Tab-Switch Detection Algorithm

Features

Functions

Reference



TAGUIG CITY UNIVERSITY

